# BHANU PRASAD REDDY

## ELECTRONICS ENGINEERING PORTFOLIO

VLSI Physical Design · Embedded Systems · FPGA · IoT

Engineering-minded, hands-on and curious about building reliable hardware and connected systems.

MSc Electronics Engineering with Advanced Project
University of Hertfordshire, United Kingdom

Internships • Graduate Roles • Research Opportunities

Portfolio draft • Please add your email, LinkedIn, GitHub and project evidence before sharing.

# Profile

I’m Bhanu Prasad Reddy, an Electronics Engineering postgraduate student with a background in VLSI Physical Design, embedded systems and IoT. My interests include physical-design implementation, timing analysis, digital hardware, FPGA-based design and connected systems. I’m looking for opportunities to contribute to engineering teams while continuing to develop practical implementation and verification skills.

# Technical skills

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|---|---|
| **VLSI & Physical Design** | Synopsys ICC2, PrimeTime STA, floorplanning, placement, CTS, routing, timing-analysis fundamentals, Tcl scripting. |
| **Embedded Systems** | C, Arduino, STM32CubeIDE, Proteus, microcontroller prototyping and sensor integration. |
| **FPGA & Digital Design** | VHDL fundamentals, MATLAB/Simulink, model-based design and verification coursework. |
| **IoT & Automation** | NodeMCU, Blynk, Node-RED, Raspberry Pi, connected prototypes and dashboard workflows. |
| **Other tools** | Linux, Python basics, IBM Cloud and engineering documentation. |

# Featured projects

## ORCA_TOP — 32-bit Multi-Voltage Processor

**VLSI Physical Design Training Project** — Worked through a 28 nm physical-design flow using Synopsys ICC2: floorplanning, port placement, voltage-area creation, power planning, macro and standard-cell placement, clock-tree synthesis and routing. Training figures provided: 73% utilization, 413 MHz target frequency, 9 metal layers and congestion below 1% at stages. Confirm final sign-off metrics before public use.

## Smart Agriculture using IoT

**Embedded / IoT prototype** — Project concept using NodeMCU, Blynk and Node-RED for sensor-based monitoring, remote visualization and automation. Add exact sensors, diagrams, test observations and code repository.

## Smart Bin

**Embedded systems project** — Arduino and Raspberry Pi-based project exploring sensor-driven waste-bin functionality. Add hardware list, implementation photos, control logic and test results.

## FPGA Model-Based Image Filter

**Current coursework — in progress** — Develop and verify an image-filter design through a model-based workflow, with attention to numerical precision and test evidence. Keep labelled in progress until implementation and verification are complete.

## Smart Home Automation with IoT Security

**IoT project** — Explores connected home automation with security considerations. Add platform details, threat considerations and demonstration evidence.

## Additional academic prototypes

**Embedded / IoT** — Other project experience includes weather reporting, health-monitoring IoT and atmospheric data-related work. Create individual project summaries when evidence is ready.

# Education & experience

**MSc Electronics Engineering with Advanced Project**
University of Hertfordshire, United Kingdom · Current

**B.E. Electronics and Communication Engineering**
Jansons Institute of Technology, Coimbatore · CGPA 8.79 (as provided)

**Diploma in Electronics and Communication Engineering**
Audisankara College of Engineering & Technology

# Practical experience

**Embedded Developer Intern — Mayagreens Services Pvt Ltd** | Jan 2024 – Jul 2024

Embedded development exposure and project implementation.

**Embedded Systems & Arduino Projects — Manfree Technologies** | Mar 2022 – Apr 2022

**Quality Department Intern — NS Instruments** | Nov 2019 – Mar 2020

**VLSI Physical Design Training — VLSIguru, Bangalore** | Jan 2024 – Sep 2024

# Opportunity interests

Engineering internships, graduate engineering roles and research collaborations related to electronics engineering, VLSI physical design, embedded systems, FPGA and IoT.

# Contact

**Email:** Add email address
**LinkedIn:** Add profile URL
**GitHub:** Add profile URL
**Portfolio website:** Publish the included website and add its URL

Before sharing: add contact details, verify education and experience dates, include project photos/diagrams and repository links, and confirm all reported project metrics.